

Flow Control Interfaces in Hardware Design

Valid-Ready Interface

- Handshake-based data transfer
- `valid`: sender has data
- `ready`: receiver can accept data
- Transfer when `valid && ready`

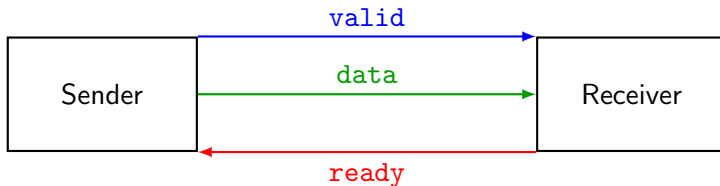
Backpressure

- Receiver controls flow using `ready`
- `ready = 0` stalls sender
- Prevents overflow and data loss

Pros: Simple, safe, widely used

Cons: Stall-dependent throughput, possible latency

Valid-Ready Interface Diagram



$$\text{Transfer} = \text{valid} \wedge \text{ready}$$

Valid–Credit Interface

- Credit-based flow control
- Receiver advertises buffer space
- Sender transmits while credits remain

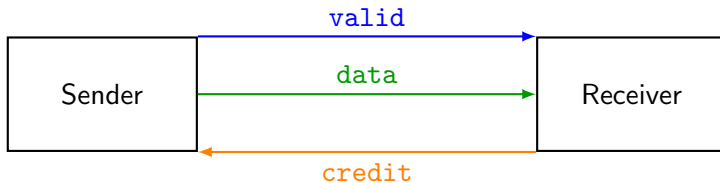
Signals

- `valid`: data sent by sender
- `credit`: available buffer slots
- Each transfer consumes one credit

Pros: High throughput, no per-cycle stall

Cons: More logic, requires buffers

Valid-Credit Interface Diagram



$$\text{Transfer} = \text{valid} \wedge (\text{credit} > 0)$$